EcoPulse Riyadh: Impact Snapshot & Roadmap

Razi Moayed Money · Industrial Engineering, Prince Sultan University · M3 Hackathon · 14 October 2026
Live tool: engrazimoney.github.io/ecopulse-riyadh · **Code:** github.com/EngRaziMoney/ecopulse-riyadh

Summary

Short version: in Riyadh, white roofs beat green roofs on energy, cost and water. The catch is that with today's cheap home electricity, even a white roof takes about 28 years to pay for itself. So getting this done is really about who pays, not about the technology.

EcoPulse Riyadh is the tool I built to show this. You pick a type of neighbourhood, choose white paint or plants, decide how many roofs to change and how much the government covers. It then tells you what happens to street heat, electricity, CO_2, cost and the owner's payback, and how it connects to the Saudi Green Initiative. I labelled every number as either sourced or my own estimate, and you can change all of them.

I worked on this alone, with help from AI for research, coding and writing. An early version of the idea was entered at NextStep Hacks 2026; the prototype, Riyadh data, deck and this roadmap were built for M3.

Problem in numbers

INDICATOR	VALUE	SOURCE
Peak land surface temperature	51.0 °C (2018) → 54.8 °C (2022)	Imam 2023
Urban heat-island area	412.8 km² (2018) → 455.6 km² (2022)	Imam 2023
Hottest / coolest land use	Industrial / vegetated	KSU Landsat study 1985–2015
AC share of home electricity, summer	About 70%	Alshahrani & Boait 2019
Grid emissions	About 0.569 kg CO_2/kWh (2021)	Climate Scorecard

One thing that stood out: the same 2023 study found that Green Riyadh's tree planting hadn't measurably cooled the city yet. Trees take years to grow. Roofs can make a difference this summer.

Impact snapshot

Scenario: 1 km² of medium-density Riyadh, 35% of land under roofs, 30% of those roofs converted (105,000 m²).

RESULT PER YEAR	COOL ROOF	GREEN ROOF
Electricity saved	1,260,000 kWh	945,000 kWh
CO_2 avoided	717 t	538 t
Street air temperature, approx.	−0.24 °C	−0.18 °C
Bill savings at 0.18 SAR/kWh	226,800 SAR	170,100 SAR
Upfront cost (one time)	6.3 million SAR	36.75 million SAR
Simple payback for the owner	**about 28 years**	**about 216 years**
Price needed for a 10-year payback	0.50 SAR/kWh	3.89 SAR/kWh
Irrigation water needed	none	yes

Neither option pays off for a household at the current tariff. So whether this happens depends on who pays. That could be a building code rule for new roofs, a rebate from the electricity company, or a city programme that starts with the hottest districts.

What it means for Saudi Arabia

If we apply the same plan to all 455.6 km² of Riyadh's hot zones (white paint on 30% of roofs), it would save about 574 million kWh and 327,000 t of CO_2 a year, 0.12% of the Saudi Green Initiative goal of 278 million t a year by 2030 (Saudipedia). Less summer air conditioning also means less crude burned for power, which reached 0.9 million barrels a day at the July peak (US EIA, 2014). If the government pays half, owners wait 14 years instead of 28.

Scalability

The tool runs on one small model with nine inputs, so it scales by changing numbers, not code.

- **Other districts:** set the area and roof share for any neighbourhood. Riyadh's heat-island area alone covers about 455 km².
- **Other cities:** swap in the local tariff, grid emissions and cooling demand. Jeddah, Dammam and other GCC cities fit the same hot-arid model.
- **Other interventions:** cool pavements and tree shade can be added using the same inputs-with-sources pattern.
- **Other users:** the same tool serves city planners, building owners and students, with built-in pages that explain how to use it in plain language.

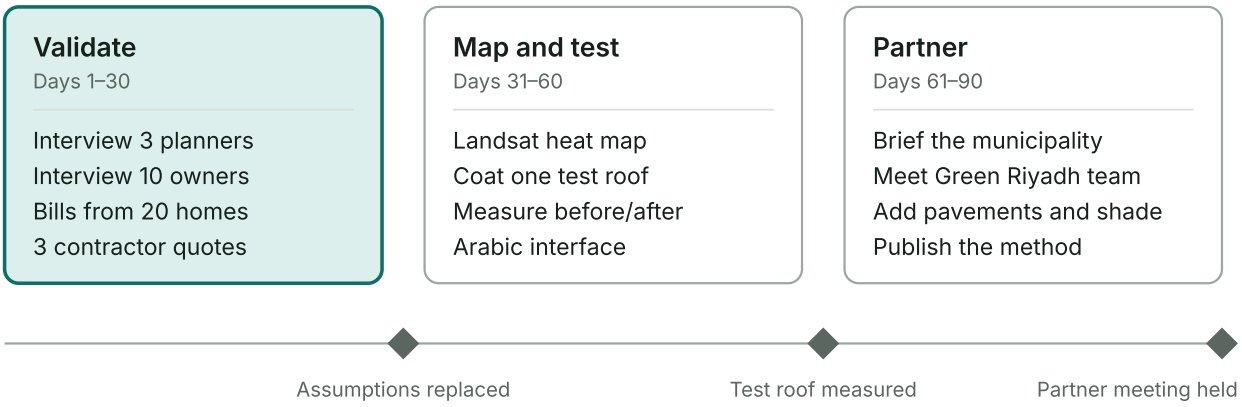
Assumptions and limits

INPUT	VALUE	TYPE
Cooling energy cut: cool / green roof	10% / 7.5%	Sourced (Elnabawi & Saber 2023)
Street air cooling at full conversion: cool / green	0.8 °C / 0.6 °C	Sourced (same study)
Household tariff	0.18 SAR/kWh	Sourced (SEC tariff)
Grid emissions	0.569 kg CO_2/kWh	Sourced (Climate Scorecard)
Cooling electricity per m² of roof	120 kWh/m²/yr	My assumption
Roof share by neighbourhood	25–50%	My assumption
Cost: cool / green roof	60 / 350 SAR/m²	My assumption

To be clear about its limits: this is a screening tool for comparing options, not an engineering design. I scale results in a straight line from one simulation study, and real cooling also depends on building height, insulation and wind. That's why the first step in my plan is replacing my three estimates with real data.

30-60-90 day roadmap

The first month is all about replacing my three biggest guesses with real Riyadh data. Nothing gets scaled up until that's done.



The test roof is the first real measurement: I'd coat one roof and compare its surface and indoor temperatures with an untreated roof next to it.

Sources

1. Imam, A. (2023). Examining the impact of Green Riyadh Project on alleviating urban heat island effects. J. Umm Al-Qura Univ. Eng. Archit., 14(4), 201–211. doi:10.1007/s43995-023-00030-4
2. Land use and land surface temperature in Riyadh, 1985–2015. ISPRS Archives XLII-3/W2, 2017. isprs-archives.copernicus.org
3. Elnabawi, M. H. & Saber, E. (2023). A numerical study of cool and green roof strategies in a hot arid climate. J. Building Performance Simulation, 16(1), 72–89. research.uaeu.ac.ae
4. Alshahrani, J. & Boait, P. (2019). Reducing high energy demand associated with air-conditioning needs in Saudi Arabia. Energies, 12(1), 87. mdpi.com
5. Saudi Electricity Company emissions intensity. Climate Scorecard, 2023. climatescorecard.org
6. Saudi Arabia electricity tariff schedule. ESMAP RISE, 2022. rise.esmap.org